

The Endocrine System

The body's second messaging service. Slower than nerves, but it reaches everywhere and it lasts.

BY THE END OF THIS PAGE YOU SHOULD BE ABLE TO

- Compare nerves and hormones on speed, reach, and how long they last
- Explain why a hormone in the blood only affects certain cells
- Say why steroid and peptide hormones act on different parts of the cell
- Trace a hormone chain from brain to gland, and predict what feedback does
- Name the major glands and what too much or too little of each looks like

1 Two messaging systems, one body

	NERVES	HORMONES
Message travels by	An electrical signal along a nerve	A chemical in the bloodstream
Speed	Milliseconds	Seconds to hours
Reach	Wherever the nerve ends	Everywhere blood goes
How long	Brief	Sustained
Good for	Pulling your hand off a stove	Growth, metabolism, reproduction, fluid balance

The two overlap on purpose. The sympathetic nerves fire in a second and are done in a second. So the adrenal gland adds epinephrine to the blood and holds the response open for minutes.

The fast system starts it. The slow system keeps it going.

2 Why a hormone in every vessel does not affect every cell

Delivery is universal. Response is not.

A hormone reaches every cell in the body. But only cells with the matching **receptor** can respond. Everything else sits in the signal and ignores it.

That is the whole basis of specificity. It also explains a lot of disease. A tissue can lose its receptors and stop responding, even when hormone levels are perfectly normal.

THINK OF IT THIS WAY

The hormone is an announcement over the whole building. Only people holding the right badge are supposed to act. Take away the badges and the announcement changes nothing, however loud it gets.

Where the receptor sits depends on the hormone. Steroid hormones dissolve in fat, so they pass straight through the membrane and bind receptors *inside* the cell. They change which genes are read, so they are slow to start and long to last.

Peptide hormones cannot cross the membrane, so they bind receptors on the *surface* and set off a messenger inside. They are fast to start and short to last. Insulin is a peptide. Cortisol is a steroid. That is why one acts in minutes and the other over hours.

3 A hormone chain, and what feedback does to it

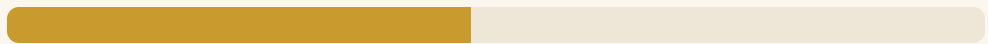
Move the hormone level and watch the whole chain respond.

HYPOTHALAMUS → PITUITARY → THYROID, WITH FEEDBACK

TRH
(hypothalamus)



TSH (pituitary)



Thyroid
hormone



Gland: normal size and activity

A normal level holds the axis in balance. Feedback is doing exactly what it does in every loop from Module 1: opposing the change and defending the set point.

- 1 **The hypothalamus** releases a starter hormone into a short local circulation.
- 2 **The pituitary** answers by releasing a stimulating hormone into the bloodstream.
- 3 **The target gland** — thyroid, adrenal, or gonad — releases the final hormone.
- 4 **That final hormone feeds back** and switches off the first two steps. Negative feedback again, exactly as in Module 1.

WHY THE NURSE CARES

A patient on long-term steroids has a shut-down chain. The pituitary stopped signaling, and the adrenal gland shrank. Stop the drug suddenly and there is no hormone from either source. That is why steroids are tapered, never stopped all at once.

4 The glands worth knowing cold

GLAND	MAIN HORMONES	TOO LITTLE OR TOO MUCH
Anterior pituitary	Growth hormone, TSH, ACTH, FSH and LH, prolactin	Several chains fail at once
Posterior pituitary	ADH and oxytocin, made in the hypothalamus and stored here	Too little ADH: huge amounts of pale urine, and constant thirst
Thyroid	Thyroid hormone, which sets metabolic rate	Low: cold, tired, slow, weight gain. High: hot, anxious, fast, weight loss
Parathyroid	Parathyroid hormone	Controls blood calcium. See Module 7
Adrenal cortex	Cortisol and aldosterone	Stress response, plus sodium, water, and blood pressure
Adrenal medulla	Epinephrine	Keeps the fight-or-flight response going
Pancreas	Insulin and glucagon	Blood sugar, in both directions

5 Insulin and glucagon — the clearest loop in the body

Two hormones from one organ, pulling opposite ways on one value.

After a meal, blood sugar rises and the pancreas releases **insulin**. Insulin opens the door for sugar to enter muscle and fat, and tells the liver to store the extra. Blood sugar falls back to normal.

Between meals, blood sugar falls and the pancreas releases **glucagon** instead. The liver breaks down its stores and puts sugar back into the blood. Blood sugar rises to normal.

Now think about what happens with no insulin, or with insulin that is ignored. Sugar stays in the blood and cannot get into the cells that need it.

So blood sugar is high while the cells act starved. That contradiction — sugar everywhere, fuel nowhere — is the core of diabetes. It explains the thirst, the urinating, and the weight loss all at once.

6 Check yourself

Five questions. The wrong answers explain themselves, so they are worth reading.

1. A hormone reaches every cell in the body but only affects a few. Why?

- a) Only some cells get blood
- b) Only cells with the matching receptor can respond
- c) The hormone breaks down before reaching most tissues
- d) Other cells make an opposite hormone

2. Cortisol is a steroid hormone. Where is its receptor?

- a) On the cell surface
- b) Inside the cell, changing which genes are read
- c) In the bloodstream
- d) On the outside of the mitochondria

3. A patient takes thyroid hormone tablets. What happens to their own TSH?

- a) It rises to match the extra hormone
- b) It falls, because feedback shuts the pituitary down
- c) It does not change
- d) It rises, and the gland grows

4. Why are long-term steroids tapered instead of stopped all at once?

- a) The drug builds up and must clear slowly
- b) The shut-down adrenal gland cannot start producing again right away
- c) Stopping suddenly causes an allergic reaction
- d) The kidney needs time to adjust

5. In untreated diabetes, blood sugar is high while the cells act starved. Why?

- a) The liver stopped making sugar
- b) Sugar cannot get into most cells without insulin
- c) Cells switched to using only fat
- d) Sugar is destroyed in the blood

Answer key

1. **Only cells with the matching receptor can respond**

Correct. Receptors decide who responds, not delivery.

2. **Inside the cell, changing which genes are read**

Right. Fat-soluble hormones cross the membrane and act on genes. Slower to start, longer to last.

3. **It falls, because feedback shuts the pituitary down**

Correct. The pituitary cannot tell a tablet from the gland. It only reads the level.

4. **The shut-down adrenal gland cannot start producing again right away**

Right. Months of suppression leave a shrunken gland and a quiet pituitary, so the patient is left with neither source.

5. **Sugar cannot get into most cells without insulin**

Correct. Fuel is there but locked out, which is why thirst, urinating, and weight loss appear together.

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